BIF401P Quiz file | Lab Quiz | 2023 Spring | Grand Quiz Marks 20 | Written by Waleed

1. The human genome consists of …… base pairs.
    - 2.3 billion
    - 2.3 million
    - **3.2 billion**
    - 3.2 million
2. In an alignment all sequences have ..... length.
    - Equal
    - Different
    - **None of these**
    - Multiple of 2
3. Sequence Alignment helps scientists.
    - To infer the functions of newly synthesized genes
    - To trace out evolutionary relationship
    - To predict new members of gene families
    - **All of these**
4. PDB allows to search following 3D structures ……..
    - **All of these**
    - Computed Structure Models
    - Experimentally determined.
    - CSM
5. GMQE (Global Model Quality Estimation) is expressed as a number between _____
    - Greater than 1
    - 0.5 to 1
    - Less than 0
    - **0 and 1**

6. Which of the following statements about the global alignment algorithm is incorrect? (Recall the concepts of Needleman-Wunsch Algorithm)
    - **Score can be negative**
    - In initialization step, the first row and first column are subject to gap penalty
    - First row and First column are set to Zero "0"
    - In trace-back step, beginning is with the cell at the lower right of the matrix, and it ends at top left cell
7. Amino acid composition denotes ..... of individual residues of a protein.
    - **Charge**
    - Strength
    - Concentration
    - Total number
8. BLASTX searches a protein sequence database using protein sequences as queries?
    - **FALSE**
    - TRUE
9. What is the main function of UniProt Archive (UniParc)?
    - To store metagenomic and environmental sequences
    - To maintain a vast collection of bioinformatics databases and services
    - **To capture all publicly available protein sequence data and contains all the protein sequences from the main publicly available protein sequence databases**
    - To provide a reference set of clusters for all UniProtKB sequences
10. TM-score _____ means a randomly selected protein pair with the gapless alignment taken from PDB
    - **<0.15**
    - <0.17
    - <0.18
    - <0.14

11. Which of the following statements about the global alignment algorithm is incorrect? (Recall the concepts of Needleman-Wunsch Algorithm)
    - In trace-back step, beginning is with the cell at the lower right of the matrix, and it ends at top left cell
    - First row and First column are set to Zero "0"
    - **Score can be negative**
    - In initialization step, the first row and first column are subject to gap penalty
12. In Target–template alignment, If the user knows the structure of the template to use for modelling, the coordinates can be uploaded in PDB format (*) together with the target protein sequence.
    - **True**
    - False
13. Which of the following statement is not true about BLAST program?
    - The programs are organized based on the type of nucleotide sequences OR nucleotide sequence to be translated
    - **BLAST is not based on heuristic searching methods**
    - The BLAST web server has been designed in such a way to simplify the task of program selection
    - The programs are organized based on the type of query sequence
14. _____ a new fold recognition method aimed particularly at detecting superfamily relationships in case of psipred
    - **GenTHREADER**
    - MEMSAT 2
    - BLAST
    - Uniprot
15. How does the sequence alignment help in drug discovery?
    - **By identifying area that might be affected by mutations or drug binding**
    - By identifying functional domains in proteins
    - **All of these**
    - By identifying evolutionary relationship between organisms

16. A protein having instability index ..... 40 is considered as stable.
    - None of these
    - Greater than
    - **Smaller than**
    - Equal to
17. _____ is a recently proposed scale for measuring the structural similarity between two structures
    - L-score
    - C-score
    - **TM-score**
    - Z-score
18. What is the main function of UniProt Archive (UniParc)?
    - To provide compressive protein sequence and functional information
    - **To provide a reference set of clusters for all UniProtKB sequences**
    - To capture all publicly available protein sequence data and contains all the protein sequences from the main publicly available protein sequence databases
    - To maintain a vast collection of bioinformatics database and services
19. Command line versions of the ClustalW programs are available for which platform?
    - UNIX
    - Window base PCs
    - Macintosh
    - **All of these**
20. Amino acid composition denotes ..... of individual residues of a protein.
    - Concentration
    - **Strength**
    - Total number
    - Charge